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# Polyphagous arthropod predators in cereal crops in central Sweden, 1979–1982

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## Abstract

During the years 1979 to 1982 sixteen cereal fields were surveyed using pitfall traps. Polyphagous arthropod and aphid specific predators were sorted and counted each week. The structure of the predator complex varies from field to field. The cumulative number of polyphagous predators up to aphid peak was shown to have a negative relationship with the number of aphids at peak, indicating that an abundant predator fauna can have an effect on aphid density. No such relationship was found with aphid specific predators. Temporal distributions of some predators are shown and their importance for aphid control discussed.

## 1 Introduction

During the past 10 years a wealth of literature has been published on cereal aphids in Europe and the potential and importance of their various natural enemies as control agents (Vickerman and Wratten 1979; Carter et al. 1980). Non-specific aphid predators have begun to be recognized as possibly influencing cereal aphid populations (Edwards et. al. 1979). Evidence of this potential influence and identification of the complex of arthropod predators concerned is by no means complete and the aim of this survey was to discover which arthropod predators occur in Swedish cereal crops and to what extent they may contribute to aphid control. Many recent studies pertain to the English winter wheat ecosystem where the main aphid pests are *Sitobion avenae* Fabr. and *Metopolophium dirhodum* Walker. In Sweden spring cereals are much more common than in England and the main aphid pest is *Rhopalosiphum padi* L. (Wiktelius 1982).

## 2 Materials and methods

During 1979–1982 16 fields were surveyed using pitfall traps (diameter 12.5 cm, depth 8.5 cm) containing detergent and water. Pitfall traps were emptied weekly in all fields. Placement and number of pitfalls varied in the 16 fields. In some of the fields trapping was done for survey purposes only, while in other fields the pitfall catches reported in this paper were taken from untreated control plots of 1 ha (or more). Placement in survey fields was either traps spread over the entire field or on a gradient from edge to middle to egde of field. Traps in other fields were located more towards the middle of the field or plot, with no traps at field edge. In table 1 a brief description of the trapping sites and times is given. Detailed descriptions of sites and trapping methods for several fields are given elsewhere [1979: 1 (Wallin et. al. 1981), 1979: 2 and 1979: 3 (Boström 1980), 1980: 3 (Wallenhammar 1982), 1981: 1, 1981: 2, 1982: 1, 1982: 2, 1982: 3

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Z. ang. Ent. 99 (1985), 433–442

ISSN 0044-2240 / InterCode: ZANEAE

*Table 1.* Description of surveyed fields. Placement of traps: W = traps in whole field, M = middle of field, G = gradient (traps placed across the field)

| Field | Crop | Size (ha) | Number of traps | Placement of traps | Trapping dates | Date of aphid peak | Aphid peak x/10 tillers |
|---|---|---|---|---|---|---|---|
| 1979: 1 | Barley cv Tellus | 12 | 36 | W | 11/6–17/9 | 18/7 | 7.4 |
| 1979: 2 | Oats cv Sang | 8.5 | 10 | W | 23/5–26/9 | 1/8 | 21.0 |
| 1979: 3 | Barley cv Tellus | 2.5 | 8 | W | 23/5–26/9 | 8/8 | 13.0 |
| 1980: 1 | Barley cv Tellus | 10.5 | 8 | M | 10/6–28/8 | 1/7 | 144.0 |
| 1980: 2 | Spring wheat cv Drabant | 52 | 20 | M | 10/6–28/8 | 1/7 | 619.0 |
| 1980: 3 | Winter wheat cv Holme | 10 | 6 | M | 21/5–31/8 | 16/7 | 50.0 |
| 1980: 4 | Barley cv Tellus | 8.5 | 12 | W | 29/5–1/9 | 1/7 | 131.5 |
| 1980: 5 | Barley cv Tellus | 6.5 | 8 | G | 29/5–1/9 | 1/7 | 79.5 |
| 1980: 6 | Barley cv Tellus | 20 | 3 | M | 3/6–31/8 | 24/6 | 880.0 |
| 1981: 1 | Barley cv Tellus | 6 | 5 | G | 21/5–11/9 | 13/7 | 14.0 |
| 1981: 2 | Barley cv Tellus | 2 | 5 | G | 21/5–11/9 | 7/7 | 54.0 |
| 1981: 3 | Barley cv Tellus | 37.2 | 5 | M | 27/5–5/8 | 7/7 | 29.0 |
| 1982: 1 | Barley cv Tellus | 20 | 5 | G | 26/5–24/9 | 13/7 | 336.0 |
| 1982: 2 | Barley cv Tellus | 4.3 | 5 | G | 26/5–24/9 | 6/7 | 544.0 |
| 1982: 3 | Barley cv Tellus | 10 | 5 | G | 26/5–24/9 | 13/7 | 170.0 |
| 1982: 4 | Barley cv Tellus | 39.3 | 5 | M | 25/5–3/8 | 13/7 | 798.0 |

(Wallin 1984)]. Fields 1979: 2, 1979: 3, 1980: 4, and 1980: 5 were located outside of Flen (58° 51′ N, 16° 38′ E). All other fields were located in the Uppsala area (59° 51′ N, 17° 41′ E).

Catches were sorted and Carabidae were identified to species and Staphylinidae to genus (or group). Spiders (Araneae) and harvestmen (Opiliones) were grouped as such.

Aphids were counted in all fields, once or twice (1980) a week. Peak numbers of *R. padi* and dates for their occurrence are given in table 1. A detailed description of aphid population dynamics in some fields is given in another paper (Wiktelius and Ekbom 1985). (Field numbers correspond in these two papers).

Linear regressions and correlation coefficients (r) were calculated for field size verus number of predators caught for some carabid species. The same was done for some pairs of species belonging to the same genaus in order to see if there was any relationship between them.

In order to see if there was any relationship between number of predators and number of aphids at population peak an analysis of covariance was done pairing the log number of *R. padi* at maximum occurrence with the log cumulative number of carabids, all polyphagous predators, and all aphid specific predators up to aphid peak. Each year was considered a "treatment". Regression lines were calculated for each year and slopes were tested for homogeneity. Where slopes were homogeneous a common slope was calculated for all four years and the intercepts for each year were computed. Common slopes were tested using a t-test to see if they differed from zero. Because of the variation in number of traps per field all values were transformed to "per pitfall trap" values.

## 3 Results

Tables 2a and 2b contain a list of the more abundant groups of potential polyphagous predators found and the numbers trapped per pitfall during the entire season. In table 3 the 10 most common ground beetles are listed and the percentage of each species when compared to the total carabid catch.

In table 4 the correlation coefficients for different insect species and field size are shown. Correlation coefficients for closely related species were *Bembidion lampros* and *B. quadrimaculatum*, $r = 0.78$, $df = 15$, $p<0.0004$, *Pterostichus melanarius* and *P. niger* $r = 0.12$, $df = 15$, NS, and *Trechus secalis* and *T. quadristriatus* $r = -0.12$, $df = 15$, NS.

Fig. 1 is a diagram of the log of the cumulative number of polyphagous

*Table 2a.* Number of predators per pitfall trap for the whole season

| | 1979: 1 | 1979: 2 | 1979: 3 | 1980: 1 | 1980: 2 | 1980: 3 | 1980: 4 | 1980: 5 | 1980: 6 | 1981: 1 | 1981: 2 | 1981: 3 | 1982: 1 | 1982: 2 | 1982: 3 | 1982: 4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CARABIDAE | | | | | | | | | | | | | | | | |
| *Agonum dorsale Pont.* | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 18.2 | 0.0 | 0.0 | 8.3 | 0.5 | 426.0 | 2.4 | 4.8 | 1.8 | 17.2 | 0.0 |
| *Amara* spp. | 2.2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 13.5 | 2.9 | 33.1 | 0.0 | 10.2 | 1.6 | 0.0 | 0.0 |
| *Bembidion guttula* Fabr. | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 18.2 | 5.6 | 2.6 | 1.6 |
| *B. lampros* Hbst | 5.5 | 29.7 | 108.4 | 3.4 | 0.4 | 6.8 | 9.0 | 38.8 | 2.0 | 133.0 | 186.1 | 4.0 | 123.2 | 74.8 | 103.2 | 1.4 |
| *B. quadrimaculatum* L. | 0.3 | 2.5 | 15.6 | 0.5 | 1.0 | 0.0 | 2.4 | 1.9 | 0.3 | 9.6 | 34.1 | 6.0 | 42.4 | 35.8 | 21.8 | 9.8 |
| *Clivina fossor* L. | 0.4 | 3.1 | 5.0 | 3.0 | 0.1 | 0.3 | 0.1 | 1.0 | 0.7 | 0.5 | 2.5 | 0.2 | 29.0 | 17.0 | 15.2 | 2.2 |
| *Harpalus affinis* Schrk. | 0 | 0.0 | 0.0 | 0.3 | 0.0 | 0.2 | 1.7 | 4.75 | 0.0 | 3.1 | 5.2 | 0.6 | 3.8 | 7.8 | 0.6 | 1.4 |
| *H. rufipes* Degeer | 27.1 | 10.1 | 49.0 | 106.4 | 6.7 | 9.3 | 14.8 | 40.1 | 1.0 | 34.6 | 98.9 | 17.4 | 31.6 | 25.0 | 10.4 | 3.6 |
| *Leistus ferrugineus* L. | 1.0 | 0.3 | 0.4 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.9 | 4.0 | 0.0 | 0.6 | 0.0 | 0.6 | 0.0 |
| *Loricera pilicornis* Fabr. | 0.1 | 1.7 | 3.9 | 6.0 | 0.0 | 6.2 | 3.6 | 14.8 | 1.0 | 2.3 | 13.9 | 0.0 | 4.0 | 1.4 | 3.6 | 1.2 |
| *Patrobus atrorufus* Ström | 4.3 | 12.8 | 52.4 | 7.3 | 50.5 | 0.2 | 0.4 | 39.9 | 0.3 | 11.8 | 2.7 | 0.0 | 5.4 | 2.8 | 21.6 | 0.2 |
| *Pterostichus cupreus* L. | 8.9 | 20.5 | 30.1 | 0.0 | 0.1 | 9.0 | 8.3 | 22.1 | 0.3 | 11.6 | 0.5 | 1.6 | 8.4 | 18.6 | 2.4 | 0.0 |
| *P. melanarius* Ill. | 74.0 | 0.8 | 13.3 | 4.6 | 106.5 | 96.8 | 1.4 | 98.0 | 30.0 | 7.1 | 2.1 | 99.4 | 110.4 | 17.0 | 41.8 | 56.4 |
| *P. niger* Scall. | 25.5 | 14.8 | 41.4 | 0.8 | 2.4 | 24.0 | 18.3 | 127.9 | 5.7 | 57.1 | 38.0 | 11.4 | 5.2 | 3.2 | 10.4 | 2.6 |
| *Synuchus nivalis* Ill. | 0.9 | 0.0 | 0.0 | 0.1 | 35.7 | 10.7 | 0.1 | 3.6 | 100.3 | 56.3 | 87.2 | 9.4 | 1.8 | 2.6 | 3.0 | 2.8 |
| *Trechus micros* Hbst. | 1.1 | 0.0 | 0.9 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 5.4 | 0.4 | 5.6 | 0.4 |
| *T. quadristriatus* Schrk. | 190.0 | 246.5 | 55.6 | 14.4 | 8.9 | 18.2 | 3.5 | 99.1 | 3.0 | 13.3 | 10.7 | 2.4 | 3.0 | 1.8 | 0.4 | 0.0 |
| *T. secalis* Payk. | 114.5 | 0.2 | 3.5 | 77.6 | 148.1 | 51.2 | 0.5 | 30.9 | 6.3 | 207.6 | 17.5 | 41.6 | 57.8 | 20.0 | 92.2 | 129.0 |

*Table 2b*. Number of predators per pitfall trap for the whole season

| | 1979: 1 | 1979: 2 | 1979: 3 | 1980: 1 | 1980: 2 | 1980: 3 | 1980: 4 | 1980: 5 | 1980: 6 | 1981: 1 | 1981: 2 | 1981: 3 | 1982: 1 | 1982: 2 | 1982: 3 | 1982: 4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| STAPHYLINIDAE | | | | | | | | | | | | | | | | |
| *Atheta* spp. (and other Myrmedoniini) | 19.7 | 48.2 | 36.6 | 29.0 | 2.2 | 0.0 | 5.9 | 76.1 | 27.7 | 122.5 | 18.3 | 28.4 | 115.6 | 70.0 | 64.8 | 72.6 |
| *Oxypoda* spp. | 0.3 | 2.0 | 24.1 | 0.8 | 0.7 | 0.0 | 0.7 | 9.4 | 1.7 | 5.0 | 29.3 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| *Oxytelus* spp. | 0.0 | 2.4 | 3.6 | 0.0 | 0.7 | 0.0 | 0.3 | 2.3 | 0.0 | 9.9 | 66.1 | 1.8 | 49.6 | 14.6 | 60.6 | 21.2 |
| *Philonthus* spp. | 0.5 | 0.6 | 6.1 | 11.0 | 0.3 | 0.0 | 1.5 | 4.3 | 0.3 | 5.8 | 33.9 | 3.2 | 29.8 | 22.4 | 15.4 | 8.6 |
| *Tachyporus* spp. | 35.6 | 3.6 | 27.0 | 1.9 | 2.3 | 3.0 | 3.2 | 4.9 | 0.3 | 5.1 | 6.6 | 2.2 | 16.4 | 5.8 | 8.6 | 14.0 |
| COCCINELIDAE | | | | | | | | | | | | | | | | |
| *Coccinella 7-punctata* L. | 8.6 | 19.8 | 2.2 | 11.8 | 4.6 | 2.2 | 20.5 | 8.5 | 9.3 | 36.6 | 35.8 | 17.8 | 38.2 | 21.4 | 13.6 | 15.8 |
| SILPHIDAE | | | | | | | | | | | | | | | | |
| *Blitophaga opaca* L. | 0.3 | 0.0 | 1.7 | 0.3 | 2.3 | 1.7 | 0.3 | 1.3 | 0.0 | 7.5 | 1.6 | 7.6 | 0.0 | 0.0 | 0.0 | 0.0 |
| ANTHICIDAE | | | | | | | | | | | | | | | | |
| *Anthicus antherinus* L. | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 8.6 | 0.0 | 3.6 | 4.1 | 25.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| ARANEAE | 36.2 | 668.4 | 454.5 | 4.5 | 5.6 | 77.5 | – | 164.8 | 4.3 | 78.0 | 41.7 | 84.8 | 144.4 | 86.8 | 73.6 | 82.0 |
| OPILIONES | 44.6 | 0.2 | 9.0 | 1.9 | 16.6 | 7.5 | – | 2.25 | 5.3 | 65.8 | 14.3 | 27.2 | 1.8 | 2.4 | 6.0 | 0.2 |

predators caught up to aphid peak occurrence verus the log of the number of aphids per 10 tiller at peak. The regression lines on the graph are as follows (y = log aphids at peak, x = log cumulative polyphagous predators): 1979, y = 3.32 − 0.86 x; 1980, y = 3.72 − 0.86 x; 1981, y = 3.43 − 0.86 x; 1982, 4.89 − 0.86 x. The slope, − 0.86, was significantly different from zero ($P < 0.0017$). The common slope for carabids was − 0.57, which also differed significantly from zero ($P<0.021$).

Fig. 2 is a diagram of the log of the cumulative number of aphid specific predators (mostly *Coccinella 7-punctata*) versus the log of aphid peak density. The regression lines on the graphs are as follows (y = log aphids at peak, x = log cumulative number of aphid specific predators): 1979, y = 1.58 − 0.35 x; 1980, y = 2.54 − 0.35 x; 1981, y = 1.96 − 0.35 x; 1982, y = 3.06 − 0.35 x. The slope coefficient, − 0.35, did not differ significantly from zero ($P < 0.3681$).

Figs. 3, 4, 5, 6, & 7 show the temporal distribution of some potential predators in several fields. The great volume of material

*Table 3.* Percent (%) in the total Carabid catch. The ten (10) most common species

| | *Agonum dorsale* | *Bembidion lampros* | *Harpalus rufipes* | *Patrobus atrorufus* | *Pterostichus cupreus* | *Pterostichus melanarius* | *Pterostichus niger* | *Synuchus nivalis* | *Trechus quadristriatus* | *Trechus secalis* | Total number of Carabids per trap for the whole season | Per trap Per week |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1979: 1 | 0 | 1 | 6 | 1 | 2 | 16 | 6 | 0.2 | 42 | 25 | 456 | 32.6 |
| 1979: 2 | 0 | 9 | 3 | 4 | 6 | 0.2 | 4 | 0 | 72 | 0.1 | 343 | 19.1 |
| 1979: 3 | 0 | 29 | 13 | 14 | 8 | 4 | 11 | 0 | 15 | 1 | 380 | 21.1 |
| 1980: 1 | 0 | 2 | 47 | 3 | 0 | 2 | 0.3 | 0.04 | 6 | 35 | 224 | 18.7 |
| 1980: 2 | 0 | 0.1 | 1.8 | 13.9 | 0.01 | 29.5 | 0.7 | 9.9 | 2.5 | 41.1 | 360 | 30.0 |
| 1980: 3 | 7 | 3 | 4 | 0.08 | 4 | 39 | 10 | 4 | 7 | 20 | 251 | 16.7 |
| 1980: 4 | 0 | 14 | 23 | 1 | 13 | 2 | 29 | 0.2 | 5 | 1 | 64 | 4.6 |
| 1980: 5 | 0 | 7 | 7 | 7 | 4 | 18 | 24 | 1 | 18 | 6 | 536 | 38.3 |
| 1980: 6 | 5 | 1 | 1 | 0.2 | 0.2 | 19 | 4 | 63 | 2 | 4 | 159 | 12.2 |
| 1981: 1 | 0.1 | 24 | 6 | 2 | 2 | 1 | 10 | 10 | 2 | 38 | 552 | 34.5 |
| 1981: 2 | 44 | 19 | 10 | 0.3 | 0.05 | 0.2 | 4 | 9 | 1 | 2 | 963 | 60.2 |
| 1981: 3 | 1 | 2 | 9 | 0 | 1 | 51 | 6 | 5 | 1 | 21 | 196 | 17.8 |
| 1982: 1 | 1 | 26 | 7 | 1 | 2 | 24 | 1 | 0.3 | 1 | 12 | 465 | 25.8 |
| 1982: 2 | 1 | 32 | 11 | 1 | 7 | 7 | 1 | 1 | 1 | 8 | 237 | 13.2 |
| 1982: 3 | 5 | 29 | 3 | 6 | 1 | 12 | 3 | 1 | 0.1 | 26 | 353 | 19.6 |
| 1982: 4 | 0 | 1 | 1 | 0.1 | 0 | 27 | 0.4 | 1 | 0 | 61 | 138 | 12.5 |

*Table 4.* Correlation Coefficients. Field size vs species

| | r | P<F |
|---|---|---|
| Araneae | 0.32 | 0.24 |
| *Atheta* spp. | 0.27 | 0.31 |
| *Bembidion lampros* | −0.497* | 0.05 |
| *Bembidion quadrimaculatum* | 0.24 | 0.37 |
| *Clivina fossor* | 0.15 | 0.59 |
| *Coccinella 7-punctata* | 0.24 | 0.38 |
| *Harpalus rufipes* | 0.44 | 0.08 |
| *Patrobus atrorufus* | 0.09 | 0.74 |
| *Pterostichus cupreus* | −0.55* | 0.03 |
| *Pterostichus melanarius* | 0.55* | 0.03 |
| *Pterostichus niger* | −0.41 | 0.11 |
| *Synuchus nivalis* | 0.02 | 0.95 |
| *Trechus quadristriatus* | 0.25 | 0.35 |
| *Trechus secalis* | 0.31 | 0.24 |

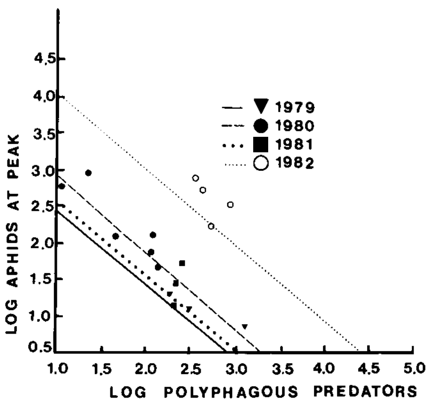
makes it impossible to present the temporal distributions for all species.

## 4 Discussion

The list of potential predators found in Swedish fields contains one interesting deviation from other lists from other European countries. *Trechus secalis* has not been recorded from arable land outside of Scandinavia. Borg (1973), Nedstam and Jonasson (1974), and Basedow et al. (1976) have reported *T. secalis* earlier from Swedish strawberry, cabbage, and cereal fields. Andersen (1982) has reported *T. secalis* from Norwegian swede and cauliflower fields and Varis et al. (1984) from cabbage, sugarbeet, and timothy in Finland. Most of the species and groups in the present list have been shown to eat aphids (Sunderland 1975; Vickerman and Sunderland 1975).

A possible artifact of the placement of the traps may be that catches of spring breeders (adult overwinterers) *Bembidion lampros* and *Pterostichus cupreus* were low in fields where no traps were placed at the edge of the fields. These were also fields of 10 ha or more. *B. lampros* and *P. cupreus* may use the field edge as a hibernation site (Wallin 1984) and if no traps were placed at the edges in a large field *B. lampros* and *P. cupreus* beetles may not have been able to disperse far enough into the field to be detected by traps in the middle of the field.

Of the 10 most common carabid species found in the fields 1–3 species seem to dominate in each field. The dominating species vary, however, from field to field. Field size might be one factor that is important in a complex of variables



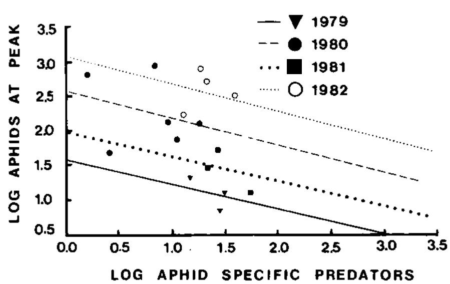




*Fig. 1 (left).* Regression between polyphagous predators and *R. padi* numbers. Each point is one field. – *Fig. 2 (right).* Regression between aphid specific predators and *R. padi* numbers. Each point is one field

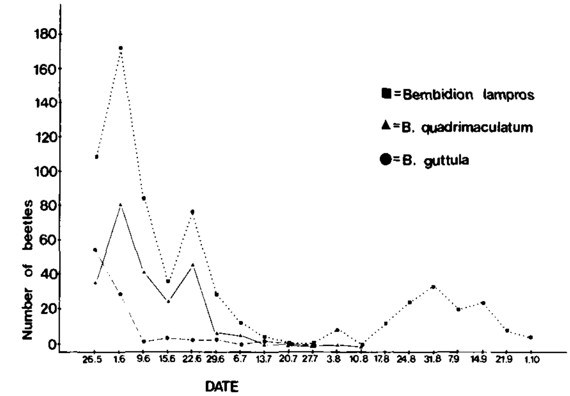



*Fig. 3.* Temporal distribution of three (3) *Bembidion* species. Numbers are the total catch for each week. Field 1982: 1

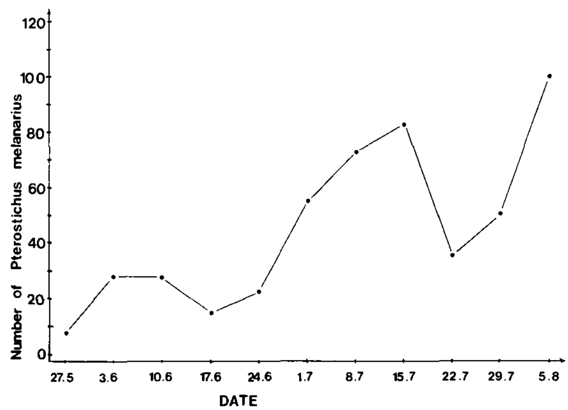



*Fig. 4.* Temporal distribution of *Pterostichus melanarius*. Numbers are the total catch for each week. Field 1981: 3

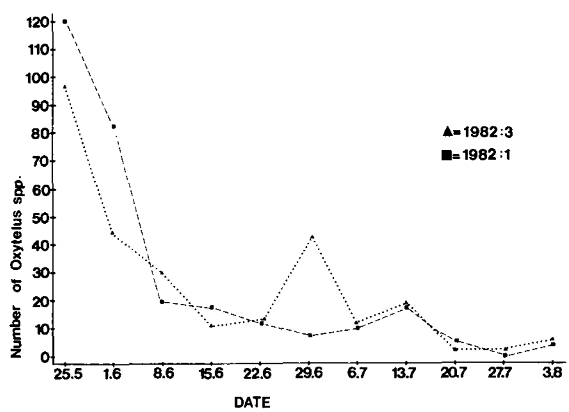



*Fig. 5.* Temporal distribution of *Oxytelus* spp. Numbers are total catch for each week

affecting the species composition in a cultivated field. *B. lampros* and *P. cupreus* were significantly, negatively correlated to field size. That is they seem to occur in smaller fields, but this may be a result of the above mentioned problem with the placement of the traps. There is a positive, significant correlation for *P. melanarius* and field size and a negative (but non significant) correlation for *P. niger*. There is, however, no significant, negative correlation

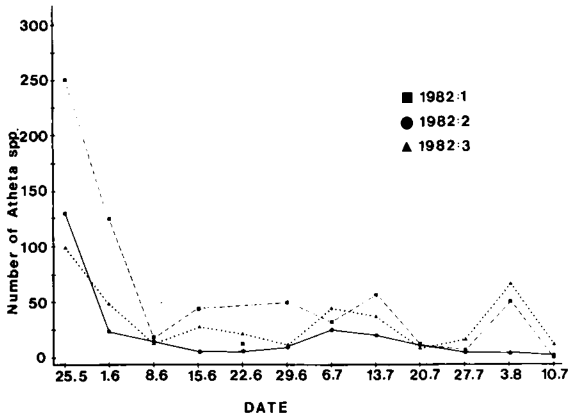



*Fig. 6.* Temporal distribution of *Atheta* spp. Numbers are total catch for each week

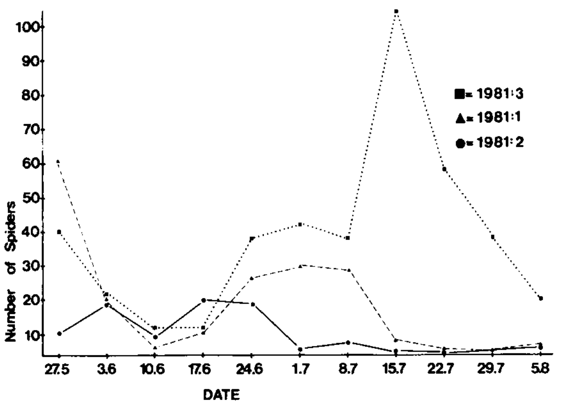



*Fig. 7.* Temporal distribution of spiders. Numbers are total catch for each week

between numbers of *P. melanarius* and *P. niger* so no evidence of a competative relationship could be seen.

The regressions shown in fig. 1 for years 1979–1981 support the hypothesis that an abundant predator fauna can have an effect on aphid population density. The different intercepts for the regression lines with a common slope indicate different levels of aphid populations. The intercept for 1982 was greater than that for 1980, a year in which aphids were also abundant. The range of aphid and predator densities for 1980 was, however, broader than in 1982. Aphid migration in 1982 began earlier than usual and was drawn out, probably due to bad weather, over two to three weeks (Wiktelius unpubl.). In general aphid population levels seemed to be higher in 1982 than in 1980. A large aphid input into a field, such as in 1982, may preclude aphid control by predators despite a relatively rich predator fauna.

The common slope for aphid specific predators did not differ statistically from zero indicating no relationship between numbers of aphid specific predators and aphid densities. The range of aphid specific predator catches was 1.5 – 52.8 per trap up to pest peak while the range of polyphagous predator catches was 11 – 1184 per trap up to aphid peak. In many years catches of *C. 7-punctata* were most abundant shortly after aphid peak density suggesting a response to aphid density. Aphid specific predators, however, most definitely add to the general predation pressure.

For a predator to be of consequence for aphid population development it must be present when the emigrants start to arrive in the field. Because *R. padi* is holocyclic in Sweden and spring cereals are being considered here a more definite starting point can be given for this system (between 15 May and 15 June) (WIKTELIUS 1982) than in winter cereals with other aphid species. For carabids the spring breeders or adult overwinterers are the most likely candidates. Fig. 3 shows that three (3) species of *Bembidion* are active in field 1982: 1 up to aphid peak on the 13th of July. Other possible predators active during aphid migration an population build-up are *Clivinia fossor, Loricera pilicornis,* and *Agonum dorsale.* However, even among the so called autumn breeders (larval overwinterers) there are potential predators of consequence. Where *P. melanarius* is abundant some adults may overwinter and be active in the spring (fig. 4). Densities of *P. melanarius* are probably much lower during the spring activity period than densities during the autumn activity period. However, there is a discernable spring activity of *P. melanarius* in fields with large catches of that beetle.

Some staphylinids may be active during and after aphid migration. In fig. 5 *Oxytelus* spp. is shown to be active in 2 fields during this period. Many *Atheta* spp. (fig. 6) are also active in late May and early June.

Spiders (fig. 7) have also been caught in the pitfall traps early in the season. Levels of spider catches vary, however, among the fields (see table 2b).

Because the structure of the predator complex varies from field to field this correlative evidence points to an effect on aphid population dynamics by a collective of predators, rather than any single species or group. Also, because the potential aphid predators are polyphagous and not specific the problem of defining the mechanisms of aphid population control is further complicated by considerations of alternative prey. Any goals of manipulating polyphagous predator communities in order to increase their effectiveness as aphid control agents can only be reached by first understanding the intricate interactions taking place in and around the cereal field.

## Acknowledgements

Many thanks to HENRIK WALLIN for technical assistance and identification of material. This study was supported by the Swedish Council for Forestry and Agricultural Research and the Swedish National Research Council.

## Zusammenfassung

*Polyphage Prädatoren in Mittelschweden, 1979–1982*

Während der Jahre 1979 bis 1982 wurden mit Hilfe von Bodenfallen Erhebungen in 16 Getreidefeldern durchgeführt. Polyphage und blattlausspezifische Prädatoren wurden jede Woche sortiert und gezählt. Die Struktur des Prädatorenkomplexes war von Feld zu Feld verschieden. Das kumulierte Vorkommen polyphager Prädatoren bis zum Maximum des Blattlausvorkommens stand in einer negativen Relation zum Blattlausmaximum, was zeigt, daß eine reiche Prädatorfauna die Blattlausdichte beeinflussen kann. Dagegen gab es bei den blattlausspezifischen Prädatoren, hauptsächlich Marienkäfern, diese Relation nicht. Die zeitlichen Verteilungen einiger Prädatoren werden gezeigt, und die Bedeutung der Verteilung für die Regulierung der Blattlauspopulation wird diskutiert.

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# Field responses of *Trypodendron* spp. (Col., Scolytidae) to different concentrations of lineatin and α-pinene

By MARIA ROSA PAIVA and K. KIESEL

## Abstract

Field responses of *Trypodendron lineatum* (Oliv.) and *T. domesticum* L. to 3 dilutions of rac. lineatin in ethanol, combined in separate experiments with different concentrations of rac. α-pinene, plus ethanol, were studied in the Black Forest. Catches of *T. lineatum* were significantly improved both with lineatin at $10^{-3}$ and $10^{-2}$ ml/ml, added to α-pinene at $10^{-2}$ or $10^{-1}$ ml/ml, in relation to lower concentrations of the terpene. For lineatin at $10^{-1}$ ml/ml, significant larger numbers of ♀♀ responded to α-pinene at $10^{-1}$ ml/ml. In general, highest catches resulted from an α-pinene to lineatin ratio of dilutions of either 1 or 10. A high α-pinene concentration, of $10^{-1}$ ml/

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Z. ang. Ent. 99 (1985), 442–448

ISSN 0044-2240 / InterCode: ZANEAE